# 08. THE HEART : SETTING UP THE DEVELOPMENTAL MOVEMENT

DURATION : 10:08 STUDY SHEET


## COURSE SUMMARY

This video explores the initial formation of the heart through a process of congestion and reorganisation of the mesoderm. The course details how a concentration of primitive angiogenic cells gives rise to a cardiac tube, with embryonic flexion aligning the aortas towards the centre. The interactions between the neural tube and the amniotic cavity are also highlighted, particularly the importance of convergence on the midline. Furthermore, the video addresses the concept of cardialisation, illustrating how the morphogenetic movements of the brain and surrounding cavities influence cardiac development and other key embryonic structures, positioning the understanding of embryologists and osteopaths in relation to biodynamics.


## THE HEART: ESTABLISHING DEVELOPMENTAL MOVEMENT

### THE INITIAL FORMATION OF THE HEART

Initially, a phase of **congestion** is linked to the reorganisation of the **mesoderm** in relation to the **neural tube**. This upward trajectory leads to a concentration of primitive **angiogenic cells** upwards.

This process leads to the formation of a **cardiac tube**, initially in the form of primitive left and right aortic tubes. The differential growth rate between the internal and external tissues causes the embryo to **flex**.

This flexing movement brings the aortas closer to the centre, where they meet to form the heart. This phenomenon begins from the **precardiac zone**, located at the level of the coccyx, before cerebral development.

Initially, two aortas develop under the influence of the neural tube, which acts as a motor, and the enormous surrounding **amniotic cavity**. The differential growth of the aortas brings them closer to the midline, in front of the **notochord**.

These two tubes gradually meet at the midline. If this convergence does not occur correctly, the specific internal nutritive tissue, essential for formation, can lead to the appearance of a **corrosion field**.

This first meeting point will structure the beginning of the heart. It is a convergence on the midline of these two levels. It is crucial to adopt an "outside-in" perspective to understand this process.

Upstream of the heart, a layer of cells, precursors of the **pericardium** and the **septum transversum**, is already stretching. These **mesenchymal** cells establish an essential relationship between the heart and the diaphragm.

## CEREBRAL COILING AND CARDIALISATION

The developing **brain** coils around the heart, while the amniotic cavity exerts a circumferential thrust. In the centre, this brings back the vessels that meet in front of the notochord.

The growth rate between the cardiac tube (pure mesoderm, both motor and organiser of growth) becomes a field of restriction in relation to the pushing amniotic cavity.

This thrust is informed by the neural tube, which receives the greatest trophic information, grows the fastest, and induces the flexing movement of the embryo. This is the system that coils around the heart.

The heart remains in place; it is the surrounding system that coils. This process involves a slowing of the growth rate and a drawing together of fluids, from the periphery to the centre.

We observe here that the movement of **cephalisation** leads to the space of **cardialisation**, which will progressively lead to **diaphragmatisation**, then to **hepatisation**.

This hepatisation will then lead to **adrenalisation**, then to **gonadisation**. This is a very interesting developmental movement, linked to the development of the amniotic cavity which will completely envelop the "B zone".

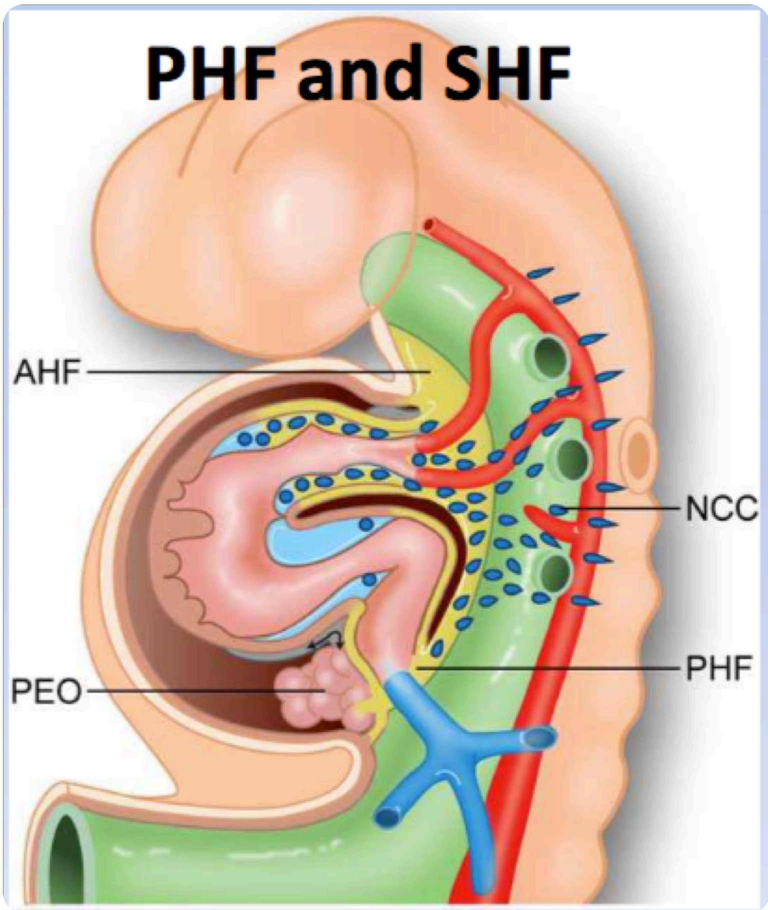


*FIG. — Embryonic heart development*

Cephalisation leads to cardialisation, which leads to diaphragmatisation, then to the entire posterior part of the peritoneal cavity.

This includes, for example, the movement of the midline in women that will form the **uterus** and **fallopian tubes**, and in men, a movement that, without going all the way, will form the **prostate** on the median plane.

Then, the **perineal** system will develop, with a return to a rebalancing of the **liver**. The liver will perform its tilting and lateral movement, becoming a major source of organisation for overall development and the closure of the anterior midline.

## MIDLINE APPROXIMATION AND THE ROLE OF THE AMNIOTIC CAVITY

It is essential to remember the **approximation** towards the midline and the initial development of the heart towards this anterior midline. Simultaneously, a significant process of cephalisation is underway.

The amniotic cavity, the developing cerebral **ectoderm**, the cardiac zone, and the diaphragmatic zone integrate. The **vitelline vesicle** progressively meets at the embryonic pedicle to form the **umbilical cord**.

All of this constitutes a developmental movement organised by the amniotic cavity. The process is a balance between the aortic system (the vascular axis) and the amniotic cavity.

The brain coils around the heart. There is a movement of integration, then a movement to form the diaphragm, the posterior parietal peritoneum, and a return around.

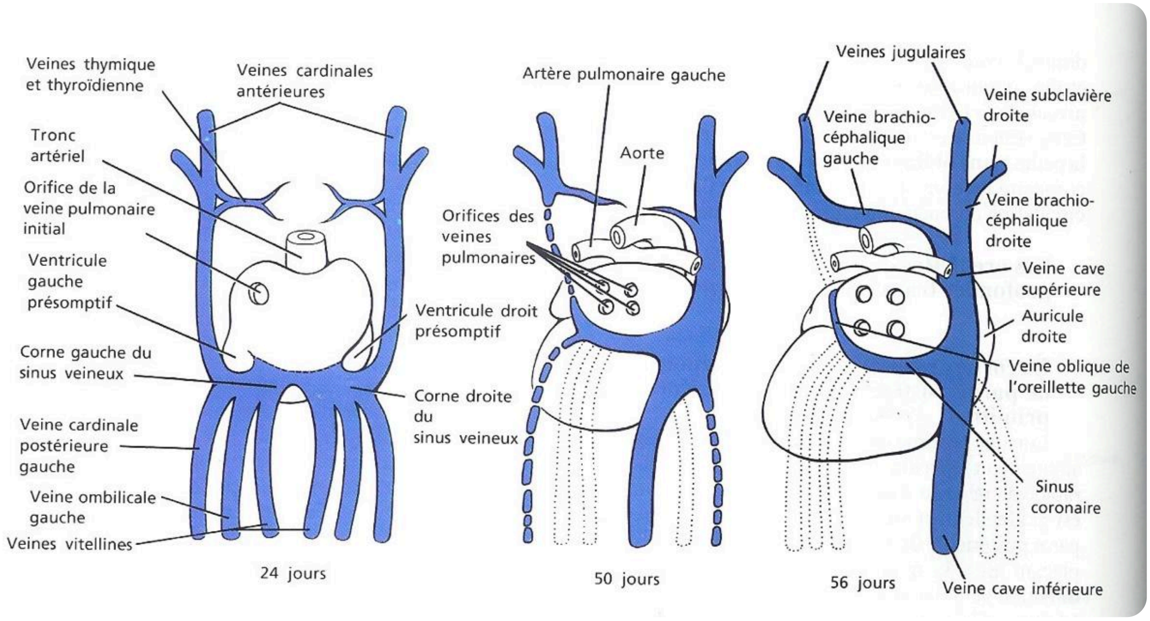


*FIG. — Cardiac venous development*

In the embryonic position, the coccyx, brain, and heart constitute points of support. The embryonic anlage depends on the **primitive streak**, which can be deviated at birth, at conception, or even pre-conceptionally.

The brain and heart are intimately linked. The heart was already present and remains in place. It is the amniotic cavity that forms around it in this movement.

## INTEGRATION OF FORCES AND THE AMNIOTIC CAVITY

Work will be done on zone B, also integrating **lateral forces**, i.e., the force of transverse thoraco-abdominal delimitation in the integration of the circulatory system, from the periphery to the centre.

Zone B, in this integration, is the amniotic cavity. A baby floats in its amniotic cavity. This membranous envelope will disappear, but its energetic aspect remains.

This "ball" around oneself is essential. The amniotic cavity forms and integrates the neural tube. In case of dysfunction (whiplash, craniosacral problem, or other), it is crucial to return to the origin.

Treat the amniotic cavity. Re-establish the spaces between the sacrum and the heart. Redo the entire movement and identify the blockages. Then, you can rework and let the system operate in the recognition of neutrals or intervene yourself.

## FROM THE AMNIOTIC CAVITY TO THE CENTRAL NERVOUS SYSTEM

A cross-section reveals the amniotic cavity above the **neural groove**. The fluid present is **primitive amniotic fluid**, which will form the neural tube. This fluid will be integrated into the tube.

The closure of the **anterior neuropore** and the **posterior neuropore** occurs around the 28th day, when the neural tube is completely closed. One must focus on zone B, because it is this that integrates the **CSF** (cerebrospinal fluid), and not the other way around.

CSF appears when the embryo implants in the mucosa. At the time of this implantation, an **exudate** appears, and it is this amniotic cavity that will create the neural tube, the neural plate, and the digestive tube.

Functionally, one cannot separate the rest of this cavity; it is always integrated. The **external coelom**, through the developmental movement of the embryo, will largely integrate into the **internal coelom**.

The rest will leave as an exudate to the outside, pushed back by the amniotic cavity. At a certain point in growth, the external coelom no longer develops.

The **neuroenteric canal** is information between the vitelline vesicle and the amniotic cavity, unrelated to the external coelom. The external coelom will be integrated into the internal coelom to form the peritoneal cavity.

However, at some point, there will be no further growth of this cavity. It seems to be pushed back, but it is actually an exudate, because the amniotic cavity grows so fast that it has integrated the internal coelom.

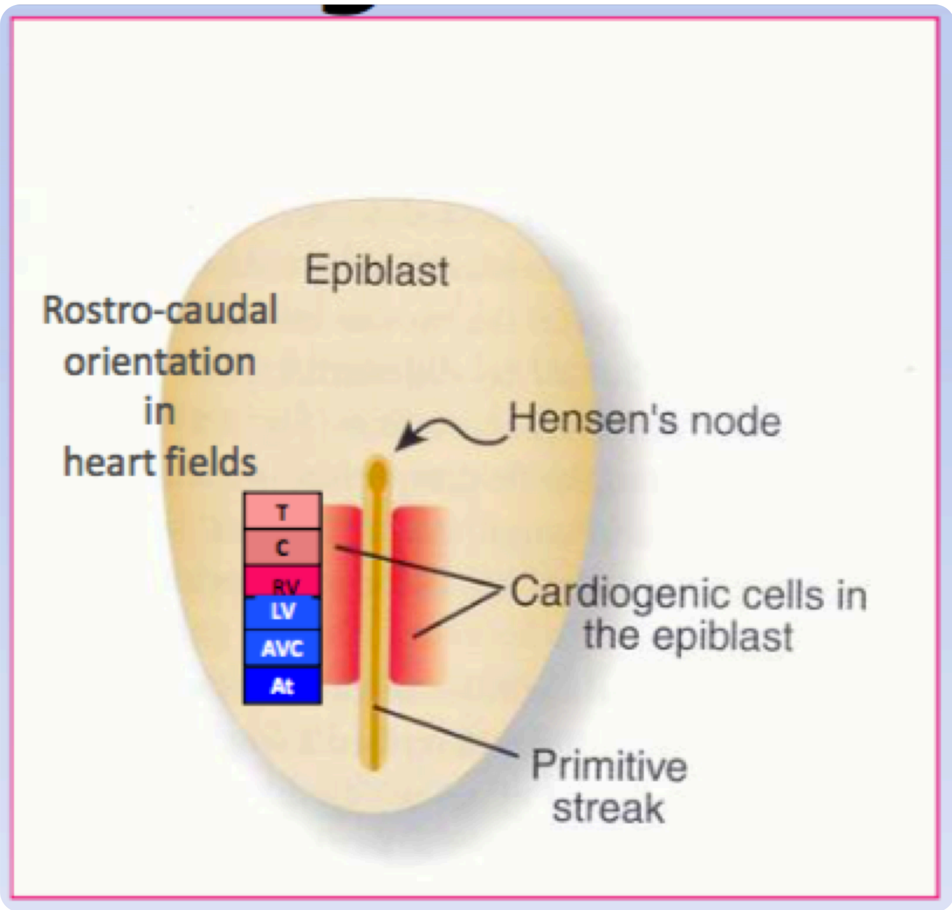


*FIG. — Epiblast, streak, heart*

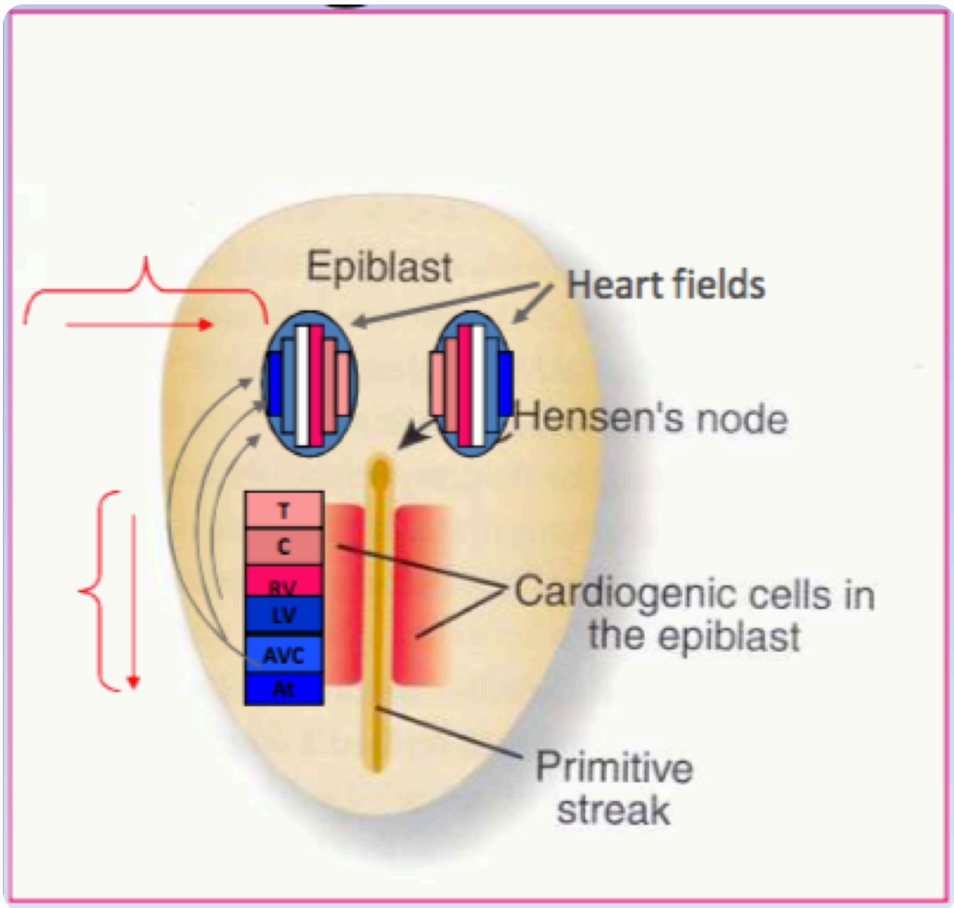


*FIG. — Cardiac cell development*

*FIG. — Early neural tube*

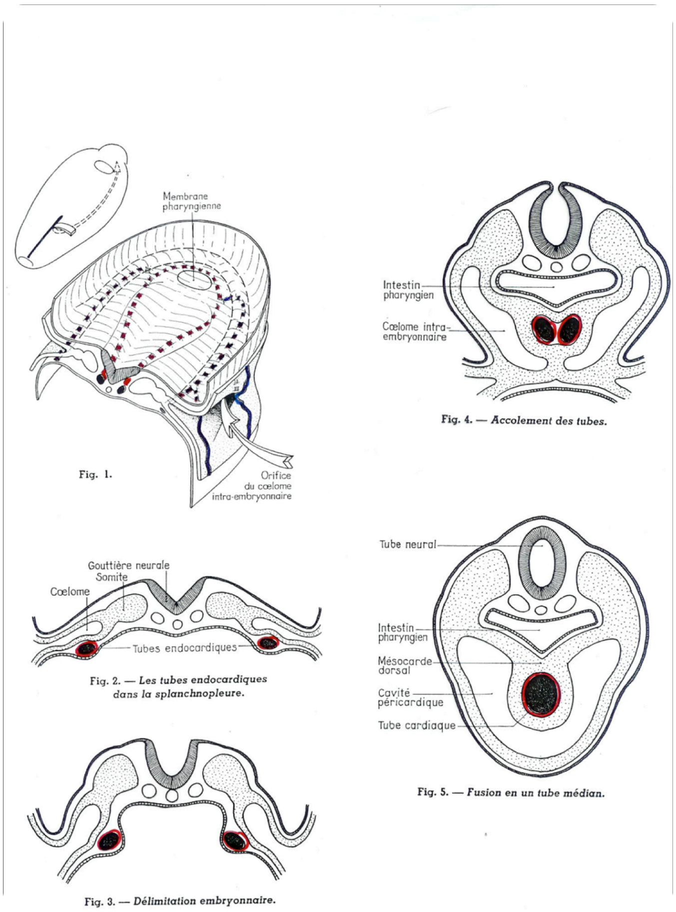


*FIG. — Embryonic heart development*

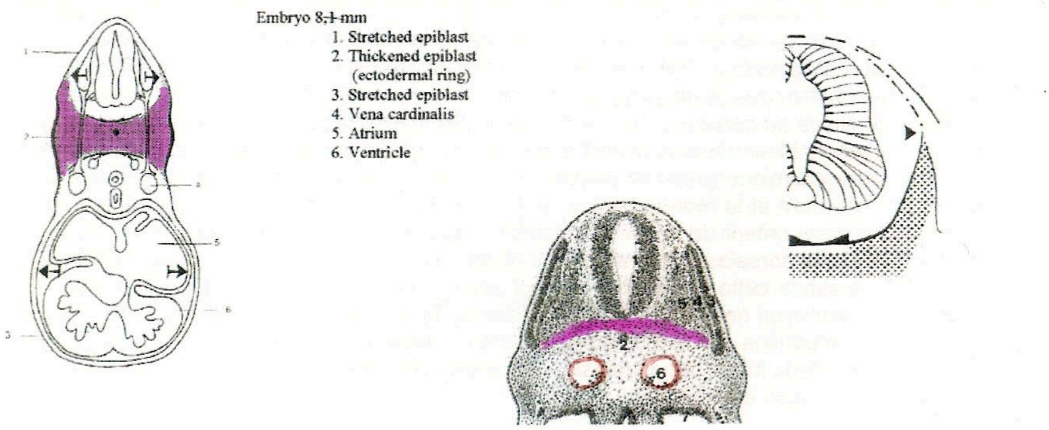


*FIG. — Heart embryonic development*